

Brandon Ho

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EDUCATION

Georgia Institute of Technology

Ph.D. in Robotics, GPA: 3.91

Masters in Computer Science, GPA: 4.00

Masters in Mechanical Engineering, GPA: 3.90

Atlanta, GA

Class of 2028

Class of 2025

Class of 2024

The Cooper Union for the Advanced of Science and Art

Bachelor in Mechanical Engineering, GPA: 3.88

New York, NY

Class of 2022

- **Minors:** Computer Science, Mathematics

RESEARCH EXPERIENCE

Warehouse Robotics Task Allocation & Path Planner, Georgia Tech

July, 2025 - Current

- Provided task allocation and path planning for a fleet of robots in Hyundai factories, improving operational efficiency by 20% and determining the optimal number of robots per scenario
- Developed warehouse robotics simulations in Unity and Isaac Sim to evaluate and optimize multi-robot performance under diverse conditions
- Used an AI agent to help operators monitor system status and receive data-driven suggestions for warehouse optimization

Multi-Drones Fire Detection and Suppression, Georgia Tech

August, 2024 - December 2024

- Constructed a multi-agent problem that focuses on effectively coordinating drones to detect and suppress wildfires, relying solely on the drones' observations under environment constraints and limited communication
- Explored various multi-drone techniques and algorithms in either detecting or suppressing wildfires in a grid-world map
- Developed a dynamic grid-world wildfire environment that incorporates realistic wildfire propagation and environmental factors, enabling robust evaluation of multi-drone cooperation strategies against multiple and unseen wildfires

Multi-Agent Dynamic Motion Planner, Georgia Tech

May, 2023 - August 2025

- Developed a generative and scalable graph motion planner in the presence of dynamic obstacles for a continuous and multi-agent setting through reinforcement learning and supervised learning on Pytorch
- Demonstrated a substantial execution speed of at least 100x faster than classical state-of-the-art motion planner (RRT* and A*) and an increase in goal-reaching & safety compared prior learned state-of-the-art motion planners (GCBF & MACBF)
- Published and presented a workshop paper titled *Towards Learning Scalable Agile Dynamic Motion Planning for Robosoccer Teams with Policy Optimization* at CoRL 2023

Heterogeneous Multi-Agents Robots, Georgia Tech Graduate Research Assistant

October, 2022 - May, 2023

- Developed a hierarchical reinforcement learning framework to accelerate policy training with simple behaviors and improve learning efficiency for complex cooperative behaviors between unmanned ground vehicles (UGVs) and unmanned aerial vehicles (UAVs)
- Enhanced reinforcement learning algorithm performance by integrating an existing library of behavior primitives, achieving superior results compared to algorithms without behavior primitives
- Conducted performance experiments in a discrete grid-world Capture-the-Flag environment using PyTorch for effective training and evaluation

Image Processing of the Sun Corona, Columbia University Laboratory Assistant

May, 2021 - August 2022

- Published a paper on understanding how coronal bright points are formed on the sun through statistics analysis
- Compiled the Hinode Satellite data into Python for image processing
- Modeled theoretical and observational distributions based on published papers on Python and IDL
- Showcased my findings and conclusion on a poster presentation

TEACHING EXPERIENCE

Machine Learning GTA, Georgia Tech Computer Science

May 2024 - August 2025

- Grade & held office hours for over 600 online graduates on the machine learning

Engineering Graphics & Design GTA, Georgia Tech Mechanical Engineering

January 2024 - May, 2024

- Grade & held office hours for 50 undergraduates on the fundamentals of engineering drawing and CADing

Math Tutor, Cooper Union ARC Tutoring

August 2021 - May, 2022

- Tutor students on Calculus I-II, Vector Calculus, Differential Equations, Probability, & Linear Algebra

Physics Teaching Assistant, Cooper Union Physics Department

August 2020 - May 2022

- Graded quizzes/Hws and set-up for physics labs
- Tutored student in Physics Mechanics and Electromagnetism and Optics & Modern Physics

Process Controls Teaching Assistant, Cooper Union Mechanical Department

August, 2021 - December, 2021

- Assisted students in completing the labs and understanding the material on feedback controls
- Maintained and tested the lab equipment with Siemens products to be in perfect capacity before the class starts

Summer STEM Teaching Assistant, Cooper Union STEM Outreach

July, 2019 - August, 2019

- Oversaw and assisted students' projects, presentation, and websites in solving the world's problems
- Formulated and taught lesson plans based on engineering process, CAD, electronics, and software

High School STEM Teaching Assistant, Brooklyn Tech STEM

July, 2017 - August, 2018

- Oversaw and assisted students in making an autonomous robot complete maze challenge
- Formulated and taught lesson plans based on how to build, program, and wire an autonomous robot called Boe-Bot Robot

WORK EXPERIENCE

Software Engineer Intern (Autonomy Control), Rivian

May, 2026 - August, 2026

- Reduced automatic emergency braking (AEB) false braking by 50% for road-model issues, improving ADAS robustness with no regression in C++
- Developed a hierarchical road-model algorithm framework to verify imminent-collision predictions across simulation and real-world driving scenarios
- Formulated perception- and sensor-derived constraints to improve the reliability of learning-based and model predictive control (MPC) models

Makerspace Operator, Cooper Union STEM Outreach

October 2019 - May 2022

- Illustrated problem-solving methods to students on electronics, microcontrollers, programming, designing, and the engineering process
- Supervised lab hours for students to work on their projects as well as 3D printing and laser-cutting
- Hosted a soldering workshop where students learned the application and importance of soldering

Electronic Hardware and Software Intern, Upcycles

May, 2020 - August, 2020

- Improved the electrical and software system for electric trikes for NYC food delivery organizations by having an organized harness and incorporating an IoT device
- Established a security system for the food delivery trikes by enabling accurate tracking of the trikes within 4 meters through GPS and a key-fob locking mechanism to prevent any unauthorized users from driving the trikes
- Lay the foundation for data streamlining of the trikes' sensor readings to the cloud through cellular communication

PROJECTS EXPERIENCE

Vision-Language Model (VLM) Planner & Verifier for Adaptive Robot Planning

Fall, 2024

- Performed at most 14% in long-horizon task success rate over existing state-of-the-art works (RT-2 and VILA) and trained learned policies with a VLM that replans a robot manipulator's trajectory with a given task prompt and current image and another VLM that verifies if its current task plan fails or changes overtime
- Optimized existing models (Gemini, LLaVA, Qwen, GPT-4) through prompt-tuning, enabling task comprehension and classification into success, partial success, or failure based on instruction and visual input
- Designed and built a novel dataset to analyze environmental perturbations' impact on long-horizon task performance
- Conducted ablation studies to assess VLM verifier configurations, demonstrating the effectiveness of a World Model with Chain-of-Thought (CoT) reasoning for robust task validation

Financial Market Prediction Using Limit Order Book Data

Fall, 2024

- Engineered 20 novel features (e.g., Volume Imbalance, RSI, Momentum, and Moving Average Crossovers) to enhance machine learning predictive performance in mid-price prediction for high-frequency financial data, improving F1 scores consistently across various prediction horizons
- Designed and implemented machine learning models, including Logistic Regression and Random Forest, demonstrating the value of manual feature engineering in financial market prediction
- Conducted ablation studies on state-of-the-art works (DeepLOB and NBoF) to evaluate our feature engineering's performance, displaying their effectiveness through manual feature integration

Fast Imitation Learning for Legged-Robots (FILL)

Spring, 2024

- Integrated a fast imitation learning framework called FILL for legged quadruped robot via behavior cloning (BC) and optimized adaptive modules with Optimal Transport-based reward systems for real-time correction in unseen environments
- Conducted experiments on MuJoCo environments, generating expert demonstrations for effective policy training and performing policy evaluations with JAX for computational efficiency and parallelism
- Performed ablation studies with various reinforcement learning algorithms (PPO & SAC) and imitation learning algorithms (BC, GAIL, & AIRL), analyzing performance-speed trade-offs for FILL

Scalable Cooperative Multi-Agents Task-Allocation Motion Planner

Spring, 2024

- Designed and implemented scalable reinforcement learning-based task allocation and scheduling frameworks for heterogeneous multi-robot systems, optimizing makespan and reward metrics, through Graph Neural Networks (GNNs)
- Evaluated advanced Graph Neural Network (GNN) architectures (HetGCN, HetGAT, and HetEdgeGAT) with Proximal Policy Optimization (PPO) algorithm for multi-robot cooperation and coordination in dynamic environments
- Achieved improved scalability and adaptability compared to traditional mixed-integer linear programming (MILP) solutions, enabling efficient task allocation under dynamic constraints
- Benchmarked GNN-based approaches against baseline models (MLPs and GRSTAPS), identifying the trade-offs between algorithmic complexity and real-time decision-making efficiency

3D Printer: Robustness to Vibration

Fall, 2023

- Implemented a Zero-Vibration (ZV) input shaping to effectively reduce residual vibration by 95% exerted of a 3D printer at resonance, improving the print quality and production speed
- Validated system dynamics through MATLAB simulations and experimental data, modeling the 3D printer as a second-order damped mass-spring system
- Designed a custom G-code transformation script for precise vibration testing, optimizing robustness against specific resonance frequencies for a Voxelab Aquila X2 3D printer

The Typing Robot

Spring, 2023

- Prototyped a functional PRRP typing robot by integrating servo motors, linear actuators, and a solenoid valve, demonstrating practical application and technical skills in robotics hardware assembly
- Designed a constrained reverse kinematic algorithm employing the Newton-Raphson method to efficiently search and determine feasible joint configurations for the robot, ensuring optimal performance within specified constraints on MATLAB
- Validated the feasibility of the robot's key actions, ensuring all movements remained within the defined operational space constraints

Senior Capstone Design Project: H2GO, Hydroelectric Robot

Fall, 2021 - Spring, 2022

- Earned 2nd place out of 17 teams in overall design in the ASME 2022 H2GO Student Design Competition for a cart that efficiently converts water potential energy into electrical energy and transports that same water to a unloading zone for points within a time limit through the use of a pelton turbine and an alternator
- Preform calculations on fluid dynamics, motor dynamics, thermodynamics, and mechanics of materials and simulate results through ANSYS APDL for FEA Analysis
- PCB designed a board that rectifies a 3 phase EMF from the alternator to a stable DC voltage distributed among all the super capacitors through power-electronics, incorporates a relay-mosfet motor controller, and supply power to all the cart electronics
- CADed/laser-cut the cart on Onshape and iterated multiple 3D-printed parts to generate the most efficient energy conversion
- Execute convex optimization schemes on Python to find the most efficient design in getting the most points and a PID cascaded velocity & yaw controller on Arduino to minimize energy consumption

Cooper Union Motorsports FSAE, Low Voltage Electronics System Lead

Fall, 2018 - Spring, 2022

- Lead and manage the Low Voltage Electronics System, consisting of 2 subsystems: Harness and Data Acquisition
- Created a serviceable and reliable wire harness by color-coding wires, incorporating connectors with a water tightness rating of at least IP-65, grouping bundle of wires together
- PCB-designed the backboard as a centralized power distributor in delivering power to all low voltage electronics and providing over-current protection to minimize points of power failures
- Collected, plotted, and analysed data from the car after testings to justify and improve the overall car design parameters
- 3D-printed, water-tighted enclosures and laser-cut mounting tabs for holding electronics on Solidworks & NX

Drone Controls of Crazyflie

Fall, 2022

- Programmed & controlled the Crazyflie flight in Python through ROS for communication and Vicon cameras as sensors
- Executed trajectory tracking for the Crazyflie to follow a remote-control car inputted by the user
- Preformed path following control for the Crazyflie to do a Figure 8, a circle, and a pendulum
- Ensured safety of the Crazyflie through obstacle avoidance algorithms

Autonomous Robot Tank: The Rat

Fall, 2021

- Designed/fabricated a 12V motorized, ping-pong shooting, autonomous robot for a robot tank battle competition
- Constructed the robot's wire schematic and algorithms on Arduino for target tracking, PID controls, and shooting mechanism
- Received the highest and only rank as Challenger for completing all the challenges in the competition
- Produced a written technical report on its design and algorithms

CFD Analysis of Axial Compressor for a Gas Turbofan Engine

Spring, 2021

- Designed a 9.45 ft long, 6 ft wide, 20 stages axial compressor with a minimum pressure ratio of 20:1 to operate at an altitude of 40,000 ft and at Mach 0.75
- Achieved results through Solidworks for CADing, Altair Hypermesh for meshing and Fluent for CFD analysis
- Justified the design and results in a written technical report and in front of engineers with CFD-related backgrounds

FEA of Wind Turbine Tower and Base Design

Fall, 2020

- Modeled a 500 ft tall wind turbine to sustain itself under its own weight of 2875 tons, a wind speed of 105 mph, and temperatures, ranging from -10°F to 120°F, under with a factor of safety of 2.3
- Gained results through Solidworks for CADing, ANSYS APDL & Workbench for meshing, and FEA structural/thermal analyses
- Justified the design and results in a written technical report and in front of engineers from construction-related industries

SKILLS

Languages: Python, C/C++, MATLAB, IDL, Pytorch, ROS2

Software : LabVIEW, Arduino, SolidWorks, Siemens NX, Onshape, ANSYS APDL/Workbench/Fluent, Altair Hypermesh, Eagle, Altium

Hands-on : Soldering, Casting, Crimping, Welding, PCB-designing

LEADERSHIP

Georgia Tech Mechanical Engineer Graduate Association Interdisciplinary VP

August, 2025 - May, 2026

- Direct engagement initiatives for a diverse cohort of 500+ mechanical engineering graduate students
- Spearhead the end-to-end planning and execution of interdisciplinary networking events, fostering collaboration between departmental sub-groups and disciplines
- Monitor a multi-event budget, ensuring fiscal responsibility and high-impact resource utilization

Georgia Tech RoboGrads Social Vice President

August, 2024 - May 2026

- Raise the morale of over 200 Georgia Tech Robotics Programs graduates
- Oversee & planned events that bring the robotics graduates together for socialization & enjoyment
- Communicate with Georgia Tech administration office over the events logistics & budget

Georgia Tech VIP XPrize UAV Macro Planning System Lead

August, 2024 - December 2025

- Oversaw the multi-drone coordination and planning in detecting and suppressing large-scale wildfires for the XPrize competition
- Constructed a simulation, incorporating sensing and suppression drones and wildfires with realistic environment dynamics
- Developed multi-drones task and motion planning algorithms (TAMP) to search for wildfires and suppress them accordingly

Georgia Tech Starter Bikes Treasurer & Mechanics

August, 2022 - Current

- Oversee the finance of bike-related equipment for facilitating over 1000 Georgia Tech cyclists
- Restored damaged/forgotten bikes to their restored conditions
- Provided hands-on instruction to students in repairing their bikes

Cooper Union Motorsports Low Voltage System Lead

August, 2021 - May, 2022

- Oversaw the Low Voltage Electronics (LVE) System to build and design an EV car for the Motorsport FSAE competition in Michigan
- Presented bi-weekly progress report of the LVE state to the rest of leadership and 30 other members
- Communicated with other system leads and members in the team to achieve our desired goals and requirements
- Provided hands-on instruction in using the electrical tools and PCB-designing software for building the car

RESEARCH INTERESTS

- Deep Learning (reinforcement learning & imitation learning)
- Multi-agent reinforcement learning
- Motion planning and navigation
- Locomotion and dynamics of hand-manipulators, multi-legged, aquatic, and aerial robots
- Control theory (Trajectory Generation & Planning)
- Numerical methods and scientific computing in Python

PUBLICATION

1. Ho.B, Rogers.N, Choi.S, (2026) Shared Roadmap Reconstruction on Non-Grid Environments for Multi-Agent Path Planning via Heterogeneous Graph Learning @ IEEE RA-L 2026 [Under Review]

2. Ho.B, Choi.S, (2026) CERMs: Learned Roadmap Sparsification for Multi-Agent Path Planning via Clustering and Latent Encoding @ ICRA 2027 [Under Review]

3. Andrade.M, Cha.J, Ho.B, Srihari.V, Yadav.K, Kira.Z, (2026) Let's Think in Two Steps: Mitigating Agreement Bias in MLLMs with Self-Grounded Verification @ ICLR 2026

4. Andrade.M, Cha.J, Ho.B, Srihari.V, Yadav.K, Kira.Z, (2025) Let's Think in Two Steps: Mitigating Agreement Bias in MLLMs with Self-Grounded Verification. Workshop on Multi-Turn Interaction in LLMs @ Neurips 2025

5. Ho. B, Altundas.B and Gombolay. M, (2023) Towards Learning Scalable Agile Dynamic Motion Planning for Robosoccer Teams with Policy Optimization. RoboLetics: Workshop on Robot Learning in Athletics@ CoRL 2023

6. Hahn. M, Ho. B, and Savin. D, (2022) Characteristics of Nanoflare Heating in a Coronal Bright Point. The Astrophysical Journal, 936, 113.

CONFERENCES & Presentation

1. Ho, B. "Towards Learning Scalable Agile Dynamic Motion Planning for Robosoccer Teams with Policy Optimization" Poster and presentation presented at CoRL; 2023, November 6-9; Atlanta, Georgia

2. Ho, B. "Determining Whether Coronal Bright Points (BP) are Heated by Nanoflares" Poster presented at the GothamFest; 2021, December 9-10; New York, New York

3. Ho, B. "Characteristics of Nanoflare Heating in a Coronal Bright Point (BP)" Poster presented at the Nevis Lab Columbia University; 2022, July 27; Ardsley-on-Hudson, New York

AWARDS AND HONORS

Thank-a-Teacher Note	May, 2024
Audience Choice for Senior Design	May, 2022
Best Interdisciplinary Excellence for Senior Design	May, 2022
William C. and Esther Hoffman Beller Prize (Merit in Engineering Studies)	May, 2022
Nicholas M. Stefano Prize (Outstanding Senior Project in Mechanical Engineering)	May, 2022
Cooper Union Half-Tuition Scholarship	August, 2018 - May, 2022
Tau Beta Pi Scholarship	May, 2021
American Council of Engineering Companies Scholarship	February, 2021